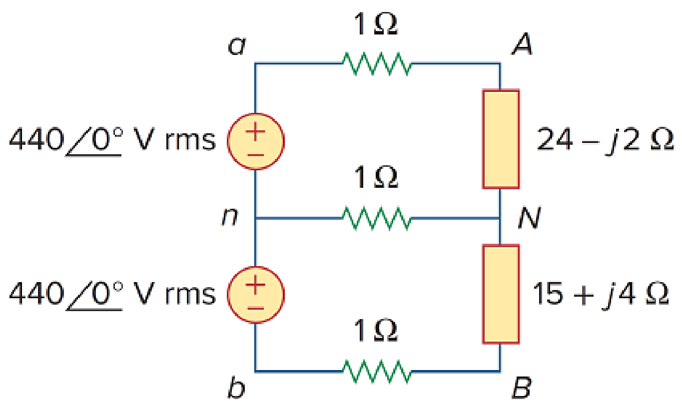
Chapter 12, Problem 86CP

Problem

For the single-phase three-wire system in Fig. 12.77, find currents **I***aA*, **I***bB*, and **I***nN*.

Figure 12.77



Step-by-step solution

Step 1 of 6

Refer to Figure 12.77 in the text book.

Identify the loops and currents in the circuit diagram shown in Figure 1.

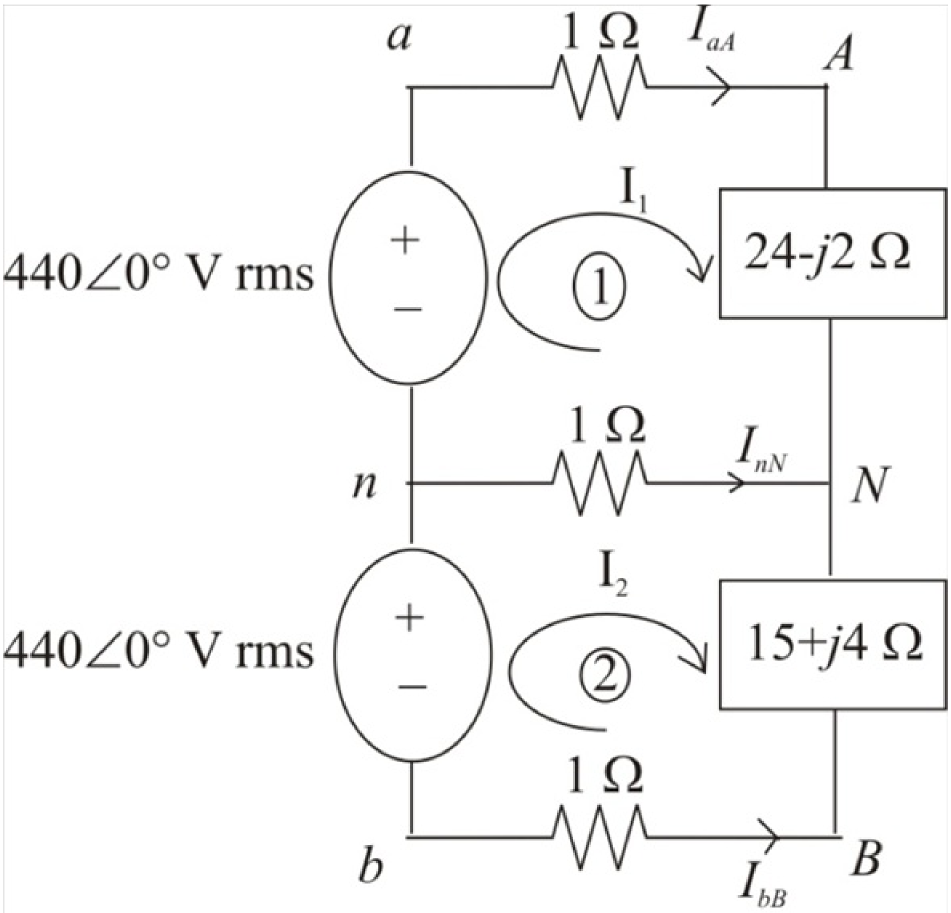


Figure 1

Step 2 of 6

Apply Kirchhoff's Voltage Law in Loop-1.

$$-440\angle 0° + I_1(1) + I_1(24 - j2) + (I_1 - I_2)(1) = 0$$
$$I_1(1 + 24 - j2 + 1) - I_2(1) = 440\angle 0°$$
$$I_1(26 - j2) - I_2 = 440\angle 0°$$
$$I_2 = 440\angle 0° + I_1(26 - j2) \;\ldots\ldots\; (1)$$

Therefore, the expression for the current I_2 is, $440\angle 0° + I_1(26 - j2)$.

Step 3 of 6

Apply Kirchhoff's Voltage Law in Loop-2.

$$-440\angle 0° + (I_2 - I_1)(1) + I_2(15 + j4) + I_2(1) = 0$$
$$I_2(1 + 15 + j4 + 1) - I_1(1) = 440\angle 0°$$
$$I_2(17 + j4) - I_1 = 440\angle 0°$$

Substitute $440\angle 0° + I_1(26 - j2)$ for I_2.

$$\left[440\angle 0° + I_1(26 - j2)\right](17 + j4) - I_1 = 440\angle 0°$$
$$\left[440\angle 0° + I_1(26.08\angle -4.4°)\right](17.46\angle 13.24°) - I_1 = 440\angle 0°$$
$$7682.4\angle 13.24° + I_1(455.36\angle 8.84°) - I_1 = 440\angle 0°$$
$$7682.4\angle 13.24° + I_1(454.37\angle 8.86°) = 440\angle 0°$$
$$I_1(454.37\angle 8.86°) = 440\angle 0° - 7682.4\angle 13.24°$$
$$I_1(454.37\angle 8.86°) = 7254.8\angle -165.96°$$
$$I_1 = \frac{7254.8\angle -165.96°}{454.37\angle 8.86°}$$
$$I_1 = 16\angle -174.82° \text{ A}$$

Therefore, the value of the current I_1 is, $16\angle -174.82° \text{ A}$.

The current I_1 represents the current I_{aA}. Therefore, the value of the current I_{aA} is,

$\boxed{16\angle -174.82° \text{ A}}$.

Step 4 of 6

Recall equation (1).

$$I_2 = 440\angle 0° + I_1(26 - j2)$$

Substitute $16\angle -174.82° \text{ A}$ for I_1.

$$\begin{aligned} I_2 &= 440\angle 0° + (16\angle -174.82°)(26 - j2) \\ &= 440\angle 0° + (16\angle -174.82°)(26.08\angle -4.4) \\ &= 440\angle 0° + 417.28\angle -179.22° \\ &= 23.46\angle -14.02° \text{ A} \end{aligned}$$

Therefore, the value of the current I_2 is $23.46\angle -14.02° \text{ A}$.

Step 5 of 6

Calculate the value of the current I_{bB}.

$$I_{bB} = -I_2$$

Substitute $23.46\angle -14.02° \text{ A}$ for I_2.

$$\begin{aligned} I_{bB} &= -23.46\angle -14.02° \\ &= 23.46\angle 165.98° \text{ A} \end{aligned}$$

Therefore, the value of the current I_{bB} is, $\boxed{23.46\angle 165.98° \text{ A}}$.

Step 6 of 6

Calculate the value of the current I_{nN}.

$$I_{nN} = I_2 - I_1$$

Substitute $23.46\angle -14.02° \text{ A}$ for I_2 and $16\angle -174.82° \text{ A}$ for I_1.

$$\begin{aligned} I_{nN} &= 23.46\angle -14.02° - 16\angle -174.82° \\ &= 38.93\angle -6.25° \text{ A} \end{aligned}$$

Therefore, the value of the current I_{nN} is, $\boxed{38.93\angle -6.25° \text{ A}}$.